Examining the Relationship Between Antarctic Penguin Species and Body Measurements*

A comparison between Univariate ANOVA and Multivariate MANOVA

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Abstract here.

*Project repository available at: https://github.com/peteragao/MATH261A-project-template.

Introduction

"Do different penguin species differ in their overall physical characteristics, and does MANOVA provide additional insight compared with conducting separate univariate ANOVAs?"

Data

The data was collected for research that was conducted as part of the Palmer Station, Antarctica, Long-Term Ecological Research program which was supported by grants through the National Science Foundation, Office of Polar Programs (NSF-OPP). Data was originally listed as three separate datasets within the Palmer Station Long-Term Ecological Research (PAL-LTER) data system and was combined into the single Palmer penguins dataset. Data were collected and made available by Dr. Kristen Gorman and the Palmer Station, Antarctica LTER, a member of the Long Term Ecological Research Network. The data used in this analysis is provided by Horst et al. (2020). All analysis was performed in R Core Team (2025).

From the provided dataset I select for the three species of Antarctic penguins, Adelie, Chinstrap, and Gentoo which will serve as the three species groups and the independent variable for my analysis. I will evaluate them based on the observed measurements of four physical characteristics bill length, bill depth, flipper length, and body mass which will be my dependent variables. Bill length, bill depth, and flipper length are recorded in units of millimeters (mm) and body mass was recorded in units of grams (g).

First we observe the dataset in Table 1. At a glance, we can see that there are some similarities between the species. Bill depth, flipper length, and body mass of Adelie and Chinstrap penguins are more similar with each other than with Gentoo. However, Chinstrap and Gentoo penguins share a similar mean bill length. Additionally, note the smaller sample size of Chinstrap penguins. If we also have unequal covariance matrices, this may affect the reliability of our MANOVA results.

Table 1: Summary Statistics of Key Variable Means and Sample Size by Species

Table 1: Summary Statistics of Penguin Measurements

species	n	mean_bill_length	mean_bill_depth	mean_flip-per_length	mean_body_mass
Adelie	151	38.791	18.346	189.954	3700.662
Chinstrap	68	48.834	18.421	195.824	3733.088
Gentoo	123	47.505	14.982	217.187	5076.016

The data is further explored in Figure 1 which shows the distribution of the four physical measurements in the dataset. From the boxplot, we see that there are a few outliers which

may affect our results. We also see some asymmetry in the body mass of Gentoo penguins and in the bill length of Chinstrap penguins.

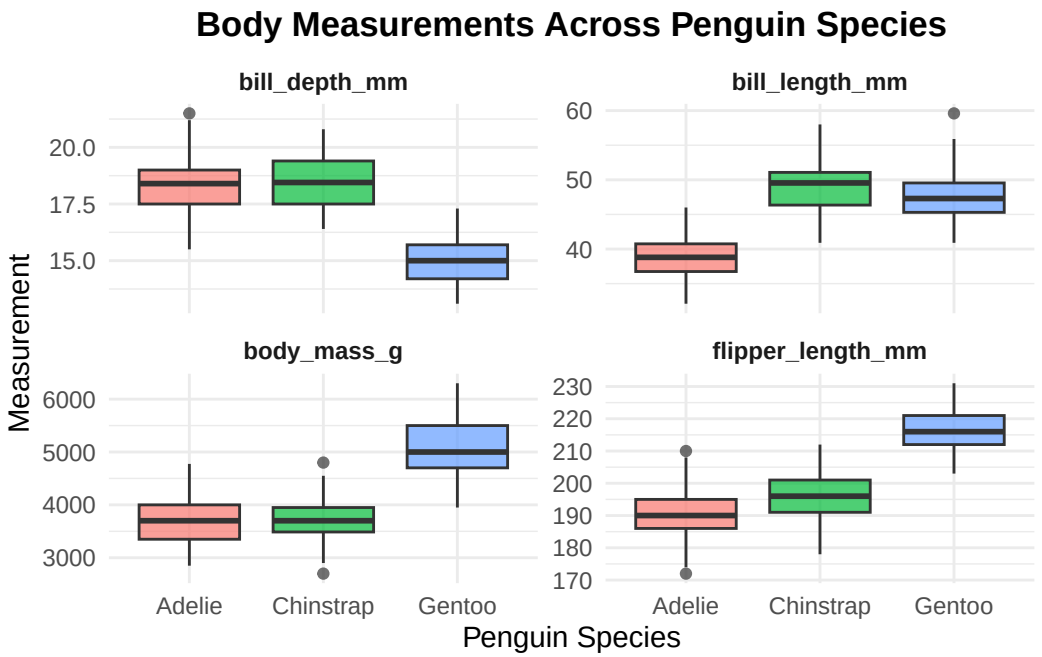


Figure 1: Visualization of Key Variable Means and Sample Size by Species

Since we expect there to be some correlation between dependent variables, I calculate the correlation matrix and present the results in Table 2 From the results, we see that bill length and bill depth are weakly negatively correlated (r = -0.235) and that flipper length and body mass are strongly correlated (r = 0.871). While flipper length and body mass are strongly correlated, there is no evidence of perfect multicollinearity.

Table 2: Correlation Matrix of Physical Measurements

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	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
bill_length_mm	1.000	-0.235	0.656	0.595
bill_depth_mm	-0.235	1.000	-0.584	-0.472
flipper_length_mm	0.656	-0.584	1.000	0.871
body_mass_g	0.595	-0.472	0.871	1.000

Methods

In order to investigate whether penguin species differ in their physical measurements I will consider four characteristics simultaneously: bill length, bill depth, flipper length, and body mass. First, I will start with four separate one-way ANOVAs tested on each of the four characteristics. The four one-way ANOVAs will test whether penguin species differ across bill length, bill depth, flipper length, and body mass simultaneously. Each ANOVA tests whether the mean of one dependent variable differs across the three species groups. The general one-way ANOVA model is

$$Y_{ij} = \mu + \alpha_i + \epsilon_{ij}$$

where Y_{ij} is the observed measurement for penguin j for species i. μ is the overall population mean. α_i is the effect of the i-th species. ϵ_{ij} is the random error associated with each observation. This error tern is assumed to be independently and normally distributed with mean zero and variance $\epsilon_{ij} \sim N(0, \sigma^2)$. For ANOVA, we assume independence of observations, normality of residuals, and homogeneous population variance.

For each of the individual univariate ANOVAs, I will have a set of hypotheses for a total of four. My Null Hypothesis for each characteristic will be

$$H_0 : \mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

with an alternative hypothesis of

$$H_A : At\ least\ one\ species\ mean\ differs$$

With Anova, we can see which specific aspect variables differ since each variable gets its own p-value. The limitation is that each ANOVA examines one outcome independently. This means we aren't directly testing whether penguin species affects the overall combination of physical characteristics. Additionally, with multiple tests, we inflate the risk of type 1 errors. Because of this, I will use MANOVA which considers all four dependent variables simultaneously. MANOVA tests whether the three penguin species have significantly different multivariate mean vectors; however, it does not identify which specific measurements differ.

The general one-way MANOVA model is

$$\mathbf{Y}_{ij} = + _i + _{ij}$$

where $\mathbf{Y}_{ij}$ is the vector of the four observed measurement for penjuin j in species i. is the overall multivariate population mean vector. $_i$ is the multivariate effect of species i. $_{ij}$ is the vector of random errors. The $_{ij}$ are independent $N_4(0, \Sigma)$ where Σ is the common covariance matrix.

Since MANOVA evaluates the combined mean vectors, we have the null hypothesis which states that all three species have equal population mean vectors

$$H_0 : \mu_{Adelie} = \mu_{Chinstrap} = \mu_{Gentoo}$$

and the alternative hypothesis

$$H_A : At\ least\ one\ mean\ vector\ differs$$

For MANOVA, we assume independence of observations, a common covariance matrix, and that each population is multivariate normal.

One assumption made when comparing two or more multivariate mean vectors is that the covariance matrices of the populations are the same. One test for equal covariance matrices is Box's M-test. The null hypothesis is that the covariance matrix for each of the populations are equal to each other and to the common covariance matrix.

$$H_0 : \Sigma_1 = \Sigma_2 = \Sigma_3 = \Sigma$$

The alternative hypothesis is that at least two of the covariance matrices are not equal. I will use this test to evaluate the common covariance matrix assumption at the end.

Results

Table 3: One-Way ANOVA Results by Penguin Species

Dependent Variable	Degrees of Freedom	F-statistic	P-value	Significance
Bill Length	2	410.600	<2e-16	***
Bill Depth	2	359.789	<2e-16	***
Flipper Length	2	594.802	<2e-16	***
Body Mass	2	343.626	<2e-16	***

If the p-value associated with Pillai's Trace is below 0.05, you reject the null hypothesis.

This indicates that the penguin species differ significantly in their combined physical characteristics.

Table 4: MANOVA Results Using Pillai's Trace

Test	Pillai's Trace	Approx. F	Numerator df	Denominator df	P-value	Significance
Pillai's Trace	1.637	379.487	8	674	<2e-16	

If the p-value associated with Pillai's Trace is below 0.05, you reject the null hypothesis.

This indicates that the penguin species differ significantly in their combined physical characteristics.

Table 5: Univariate Follow-up Tests from MANOVA

Dependent Variable	Degrees of Freedom	F-statistic	P-value	Significance
Response bill_length_mm	2	410.600	<2e-16	***
Response bill_depth_mm	2	359.789	<2e-16	***
Response flipper_length_mm	2	594.802	<2e-16	***
Response body_mass_g	2	343.626	<2e-16	***

If the p-value associated with Pillai's Trace is below 0.05, you reject the null hypothesis.

This indicates that the penguin species differ significantly in their combined physical characteristics.

Table 6: Box's M Test for Homogeneity of Covariance Matrices

	Test	Chi-square	Degrees of Freedom	P-value	Significance
Chi-Sq (approx.)	Box's M	76.795	20	1.36e-08	***

Box's M test was conducted to assess the homogeneity of variance-covariance matrices across penguin species. The test was statistically significant, $^{2}(20) = 76.795$, $p < 0.001$, indicating evidence that the covariance matrices differ across species. Therefore, the assumption of homogeneous covariance matrices was not satisfied. # Discussion

This case study demonstrates the advantages of multivariate data analysis by comparing separate univariate ANOVAs with MANOVA when examining physical differences among penguin species. While individual ANOVAs provide information about differences in specific physical measurements, MANOVA evaluates whether species differ across multiple characteristics simultaneously while accounting for relationships among the dependent variables. This provides a more comprehensive understanding of group differences and reduces the need to rely exclusively on multiple independent hypothesis tests. Therefore, multivariate analysis is particularly valuable when research questions involve several related outcome variables.

References

Horst, Allison Marie, Alison Presmanes Hill, and Kristen B Gorman. 2020. *Palmerpenguins: Palmer Archipelago (Antarctica) Penguin Data*. https://doi.org/10.5281/zenodo.3960218.

R Core Team. 2025. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. https://www.R-project.org/.